

Reproducing the SONG Paper Figures with songR

Key figures from Senanayake et al. (2021, IEEE TNNLS), reproduced with the songR R/C++ package

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2026-06-11

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About

This report reproduces the central figures of the original SONG paper —

Senanayake, D. A., Wang, W., Naik, S. H., & Halgamuge, S. (2021). Self-Organizing Nebulous Growths for Robust and Incremental Data Visualization. *IEEE TNNLS*, 32(10), 4588–4602. doi:10.1109/TNNLS.2020.3023941

— using **songR**, the SONG-inspired R/C++ implementation, on the same datasets and the paper’s Table I hyperparameters (`epsilon` = 0.99, `epochs` = 100, `a` = 1.577, `b` = 0.895) with the default UMAP dispersion. MNIST and Fashion-MNIST are subsampled for tractable runtime; COIL-20 is run in full. Cluster quality is reported as Adjusted Mutual Information (AMI; k-means on the embedding vs. ground-truth labels, mean over 5 seeds).

These figures show that **songR reproduces the qualitative structure the paper reports** — separated digit/garment clusters, COIL-20’s circular rotation manifolds, clean recovery of many Gaussian blobs, and stable incremental growth. For the quantitative *songR vs. original-Python* parity (AMI equality, Procrustes overlays), see `tests/reproduction-report/`.

Table 1: songR reproductions of the paper datasets.

Figure	Dataset	n	Clusters	AMI (songR)
Fig 4	MNIST (PCA-20)	15000	10	0.625

Figure	Dataset	n	Clusters	AMI (songR)
Fig 3	Fashion-MNIST (PCA-20)	15000	10	0.549
Fig 8	COIL-20 (PCA-300)	1440	20	0.953
Fig 7	Blobs (30c/60D)	6000	30	0.923
Fig 3/4	MNIST heterogeneous incremental	10000	10	0.462

Fig. 4 — MNIST

Handwritten digits (PCA→20). SONG separates the ten digit classes into distinct islands, as in the paper.

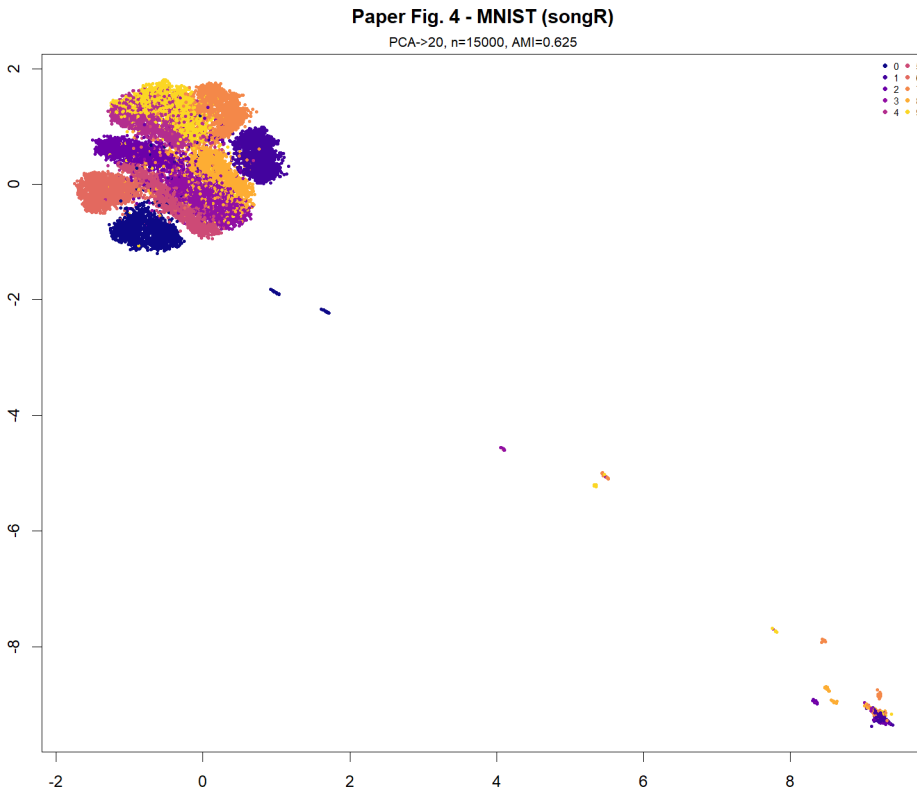


Fig. 3 — Fashion-MNIST

Garment images (PCA→20). The footwear and clothing super-groups separate, with finer structure within each — matching the paper’s harder-than-MNIST result.

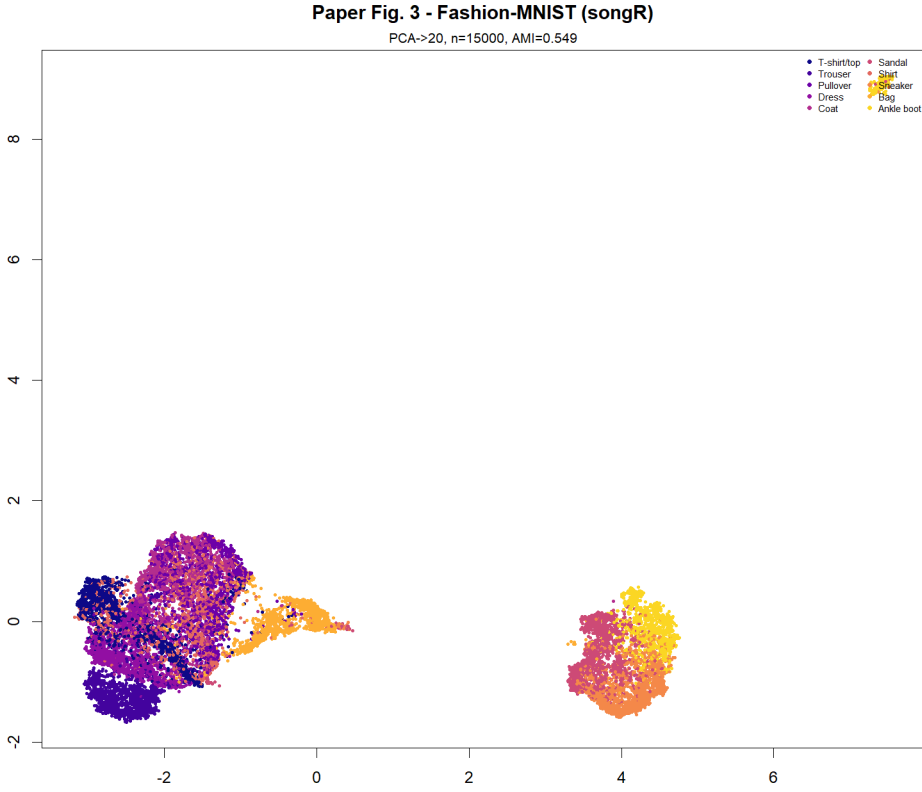


Fig. 8 — COIL-20 topology

20 objects photographed over 72 rotation angles (PCA→300). SONG preserves the rotational topology, so each object forms a **closed loop / circular cluster** — the paper’s signature topology-preservation result.

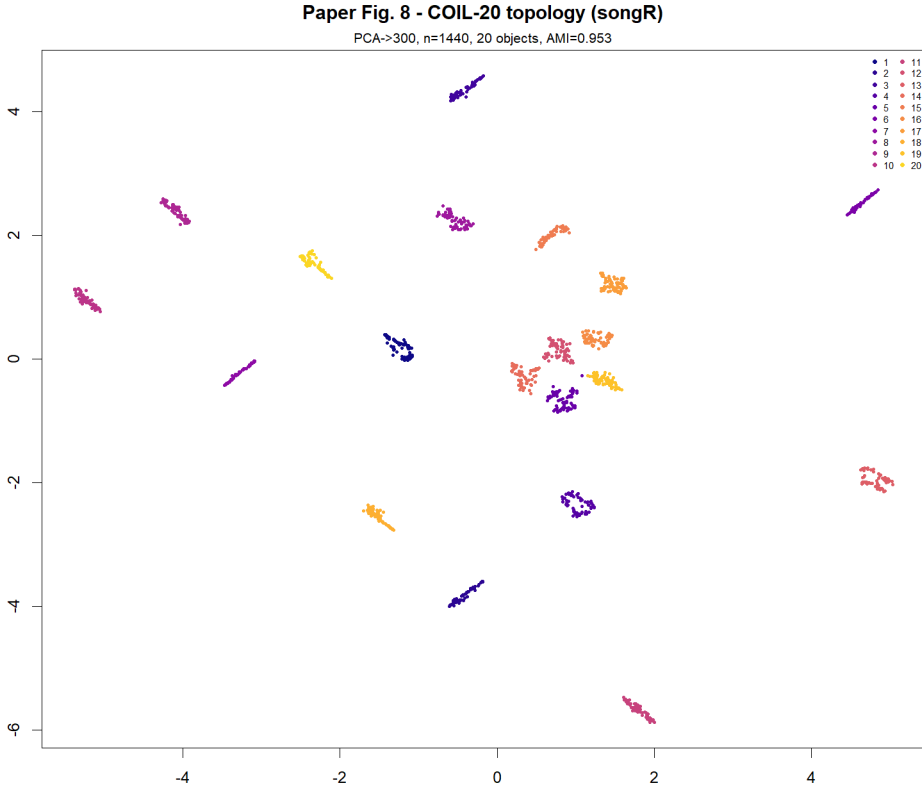
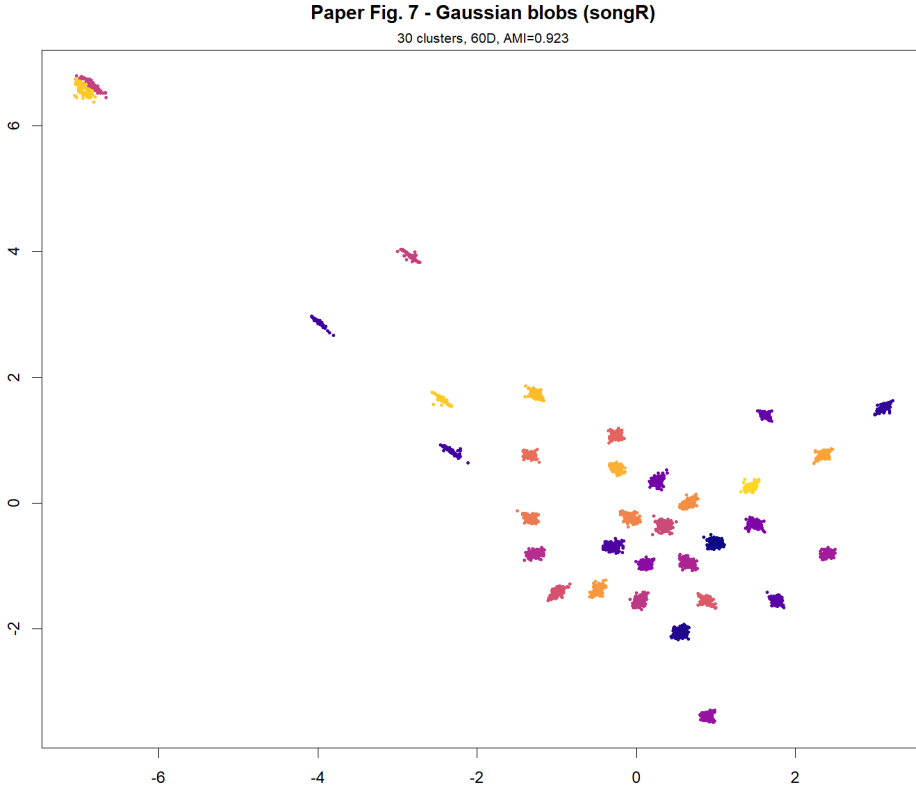


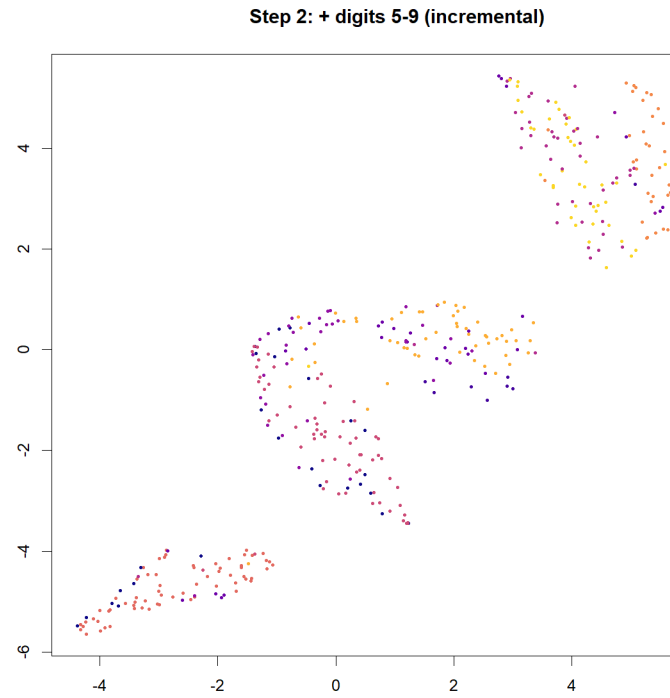
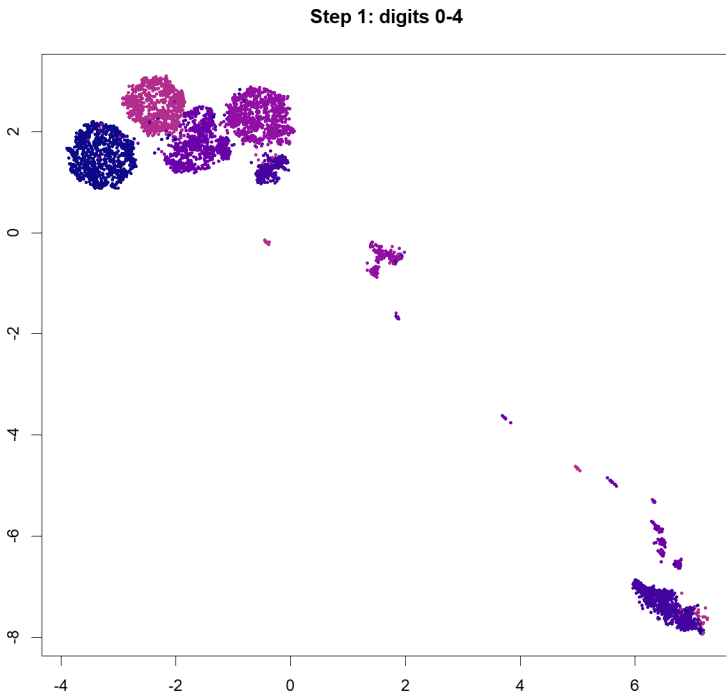
Fig. 7 — Gaussian blobs

A many-cluster, high-dimensional stress test. SONG cleanly recovers all clusters (high AMI), illustrating the paper's noise/robustness claims.



Figs. 3–4 — Heterogeneous incremental updates

SONG's signature capability: train on a subset of classes, then add **new, previously unseen** classes without reinitializing. Here we fit MNIST digits 0–4, then incrementally add 5–9 via `update()`. The original five classes keep their positions while the new ones grow into place.



Reproduce

```
Rscript tests/paper-figures/generate_paper_figures.R # figures + results.csv
quarto render tests/paper-figures/paper_figures.qmd # html + pdf
```

The real datasets are PCA-reduced per the paper protocol and kept out of the repository (`.housekeeping/`); the full-scale tutorial scripts in `tutorials/` reproduce every paper figure and table on the complete datasets.